30 Excellent!

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY
LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

ANSWER BOOK

Class No. 17 Branch S3 - MCA(2021-23)

Name DEEPAK. V Semester S3

Subject Data Science & Machine Learning

Mark Tabulation

Mark Divisions	q.1	q.2	q.3	q.4	q.5	q.6	q.7	q.8	q.9	q.10	q.11	q.12	q.13	q.14	q.15
	3	3	3	3	3	3	6	6							
Total													30		

Grand Total

PART-B

8. Classification Rules.

-> Classification Rules is a supervised learning technique which supports classification.

-> Classification Rules comprises of two major components

Classification Rules

Antecedent

Consequent

→ Antecedent

Antecedent is the component of
rules in which "if" the condition occur.
i.e. if the feature value is pro
by an example of data set.

→ Consequent

Consequent part is another compon
of the classification rule in which
th after the antecedent of "if" c
then" the class value is ~~Cate~~ Cate
by the example of data set for tra
nditions.

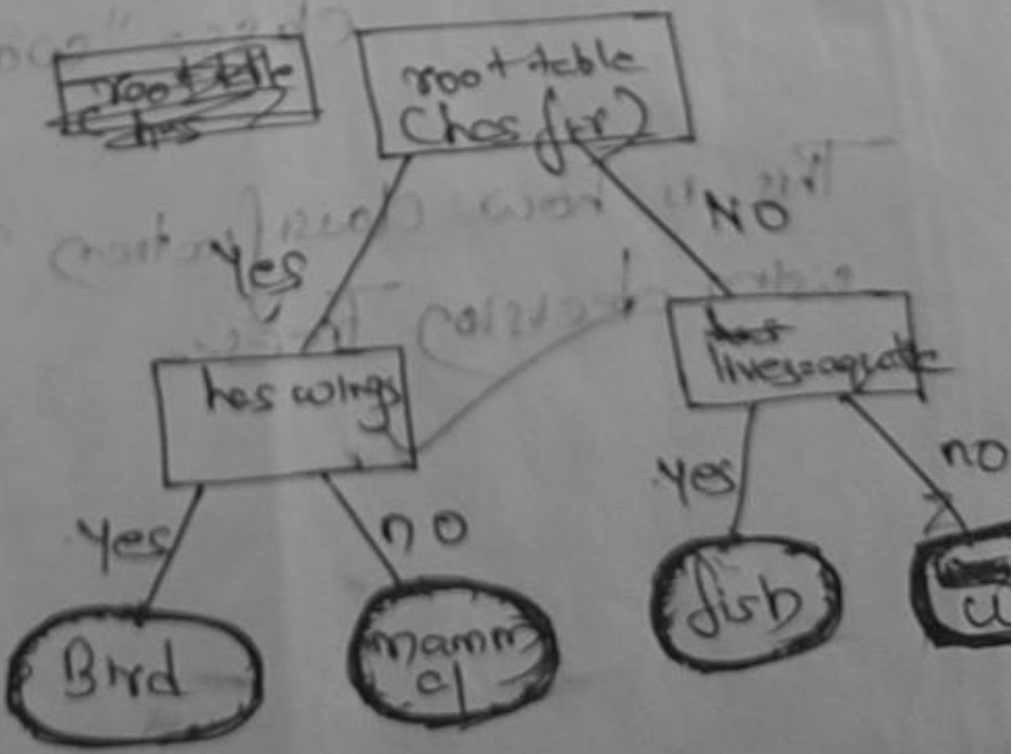
xample,

"if" has fever == "True" ←

Classification Rules Using Decision Tree

- The Classification Rules using Decision tree is obtained using "Bottom Up" approach on decision tree.
- i.e. By Looking out from the leaf node of decision tree which is categorized as one of class the class value for some example by traversing from leaf to root node we can observe the require set of classification rules for reaching to that particular classification class value.

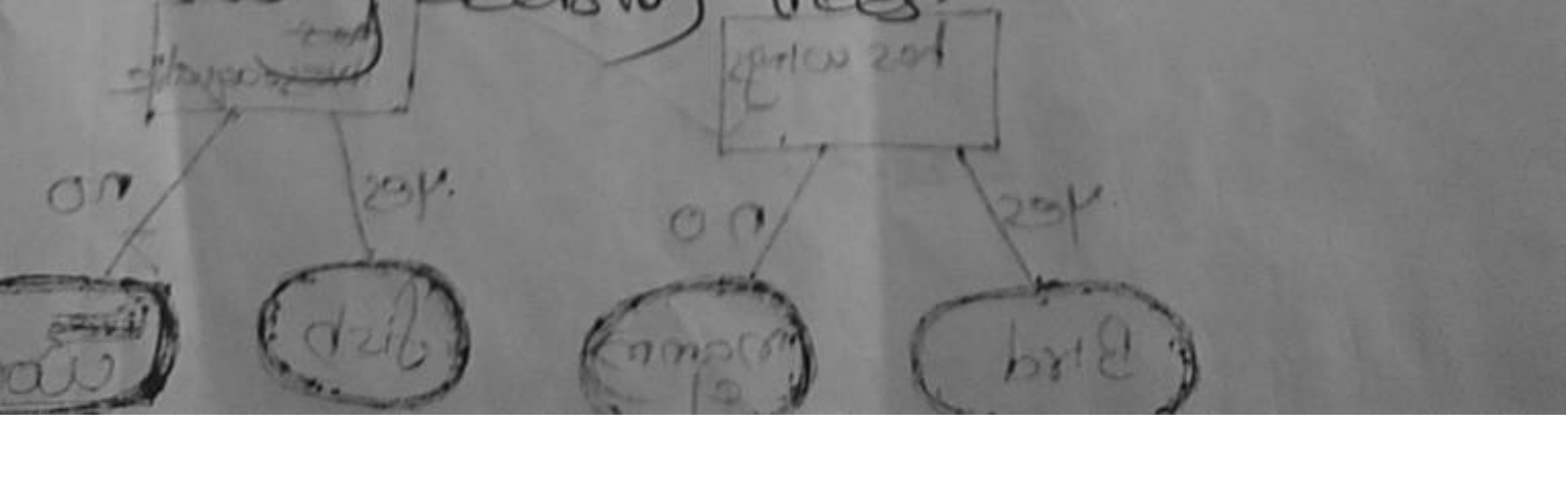
→ For example, Consider the decision tree below.



The classification rule of decision tree is:

if has_fur == "yes"
then
    if has_legs == "yes"
        class = "bird"
    else
        class = "mammal"
else
    then
        if lives == "aquatic"
            class = "fish"
        else
            class = "worm"

This is how classification rule is c
using decision Trees.



7. K-means Clustering

→ K-means Clustering is an unsupervised learning algorithm.

→ K-means Clustering mainly used for clustering or grouping of data points based on their togetherness or similarity.

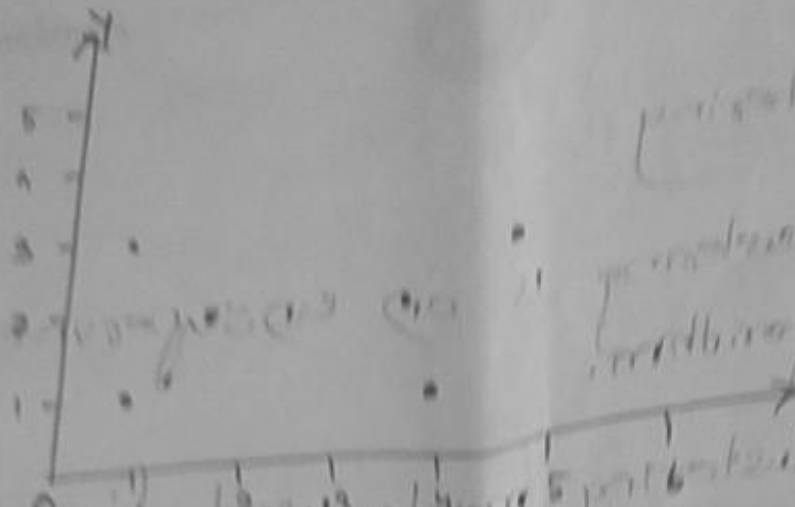
→ K-means algorithm follows an iterative approach for finding clusters.

→ Main applications of K-means is cluster formation from data points, cluster analysis etc

K-means clustering Algorithm with help of example

Consider data points,

x	1	1	2	4	4	5
y	1	3	2	1	2	3



step 1:- Consider arbitary k value
Here, considering k = 2;

step 2:- Consider k arbitary initial center points
Here, $\vec{v_1}(1, 1)$ and $\vec{v_2}(4, 1)$

step 3:- find the distance between all points with respect to center points (centeroids)

A	Distance $(1,1) = \vec{v_1}$	Distance $\vec{v_2} = (4,1)$	minimum distance	Cluster Point Class
)	0	3	0	$\vec{v_1}$
)	2	3.60	2	$\vec{v_1}$
)	1.41	2.23	[illegible]	$\vec{v_1}$
)	3	0	0	$\vec{v_2}$
)	3.16	[illegible]	[illegible]	[illegible]

The distence is found using euclediun distance

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

(1,1) (4,1)

$$\sqrt{(1-4)^2+(1-1)^2} = \sqrt{(-3)^2} = 3$$

(1,3) (1,1)

$$\sqrt{(1-1)^2+(3-1)^2} = 2$$

Step 4:- As the above table the minimum distence center points needs to be categorized cluster

step 5:- find new center points

$$\frac{1}{c_i}\sum_{i=1}^{k}(x_1c_i + x_2c_i \cdots + x_kc_i) \quad k = 1, 2 \cdots n$$

Here,

Cluster 1 = $\{x_1, x_2, x_3\} = 3$

Cluster 2 = $\{x_4, x_5, x_6\} = 3$

cluster center
New data points,

$$\vec{V_1} = \left(\frac{1+1+2}{3}, \frac{1+3+2}{3} \right)$$

$$= (1.33, 2)$$

$$\vec{V_2} = \left(\frac{4+4+5}{3}, \frac{1+2+3}{3} \right)$$

$$= (4.33, 2)$$

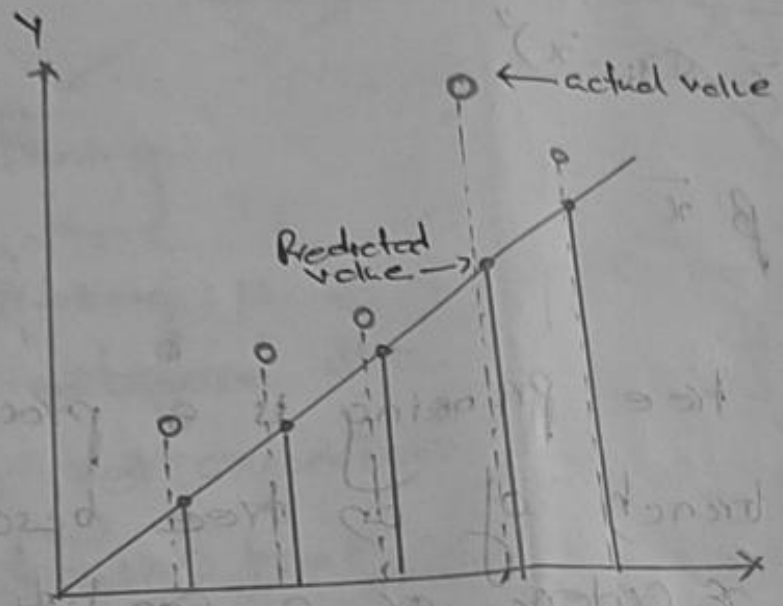
cluster center
Now new ~~data~~ points are & $\vec{V_1} = (1.33,$

and $\vec{V_2} = (4.33, 2)$

Step 6:- This process of finding new cluster center need to be done repeatedly until no new cluster center is found.

PART-A

1. Ordinary Least Square (OLS)



For data
x x_1 x_2 ...
y y_1 y_2 ...

→ OLS or Ordinary Least Square method is used for finding the minimal error between actual and predicted value

→ This is usually helps in finding the values with minimal error in linear regression

→ OLS states that

$$\sum (y_i - \hat{y}_i)^2 = \sum e^2$$

Square distance

From this,

$$\beta = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\alpha = \bar{y} - \beta \bar{x}$$

2. → The Decision tree pruning is a pr
removal of branch of an tree b
upon certain criteria or a cond

→ ~~The~~ ~~pruning~~ for a large ~~de~~ branc
decision tree. The decision tree
~~pruned~~ be overfitted or con
irrelevant classification result fo

→ for avoiding the above menti
conditions. pruning is an import
for a ~~Sustain~~ the proper exis

Decision Tree Pruning

Pre Pruning

→ The Pruning is done as before the branch get expanded on decision tree

→ Pre Pruning is done based upon

- no. of decisions which is predefined upto limit
- for small example

→ The Post Pruning

Post Pruning

→ The pruning is done after the branching of an decision

→ Post pruning is based upon

- Pruning
- which

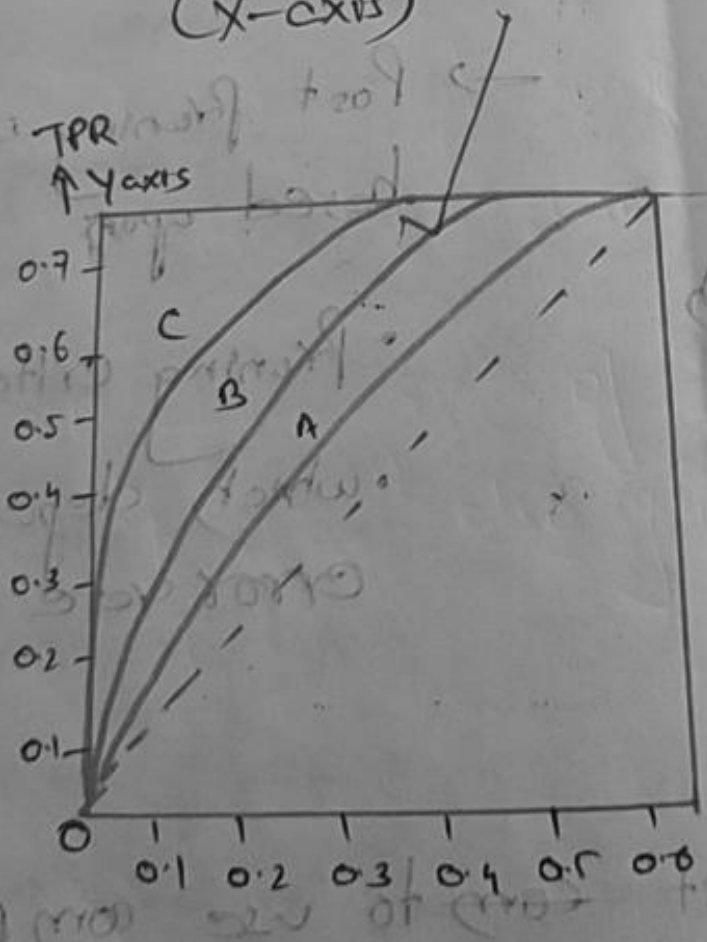
3. ROC Curve (Receiver Operating ...)

→ The ROC curve is visualizing tool performance tool for measuring the classification model performance

→ ROC having two components

True Positive Rate (TPR) (Y-axis) = $\frac{TP}{TP + FN}$

False Positive Rate (FPR) (X-axis) = $\frac{FP}{FP + TN}$



The curve which ... very close to ... is the perfect ... (these, C) ... Roc classif...

→ The area under Cure (AUC) is the ...

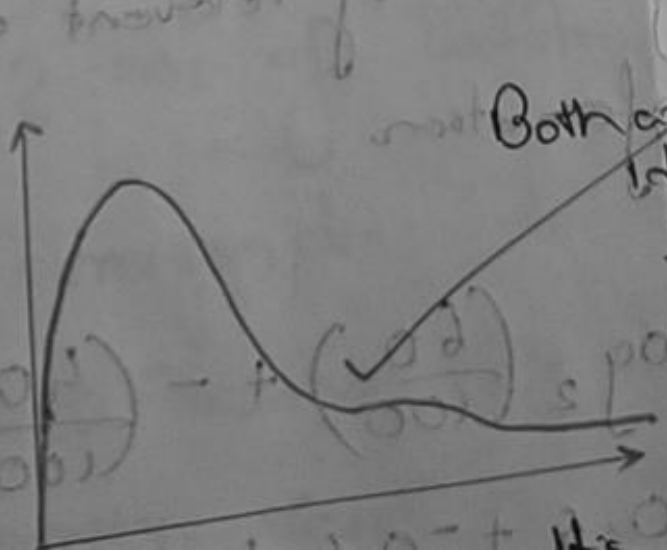
4. Boot Strap Sampling

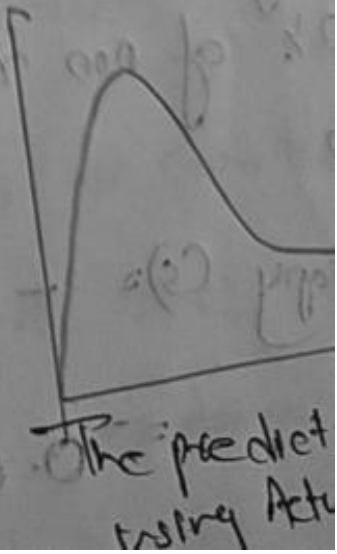
→ The Boot strap sampling is used for partitioning test and training data for model future prediction evaluation.

→ The Boot strap does samples the data in such a way that the repetation of data elements may or may not appear on a training data set.

→ This type of sampling of repeated values can minimize the use of new data elements on ~~differe~~ no: of fixed data sets and this will also generates the same results actual data

3





5. Entropy

→ Entropy is a major component to find information gain which is further used for feature extraction of nodes in decision tree

→ Entropy is used for finding the "inparity" of the data set.

→ Formula

$$\text{Entropy}(S) = \sum_{i=1}^{c} -P_i \log_2 P_i$$

where, S is the set of elements in dataset

c is no. of classes.

→ Example

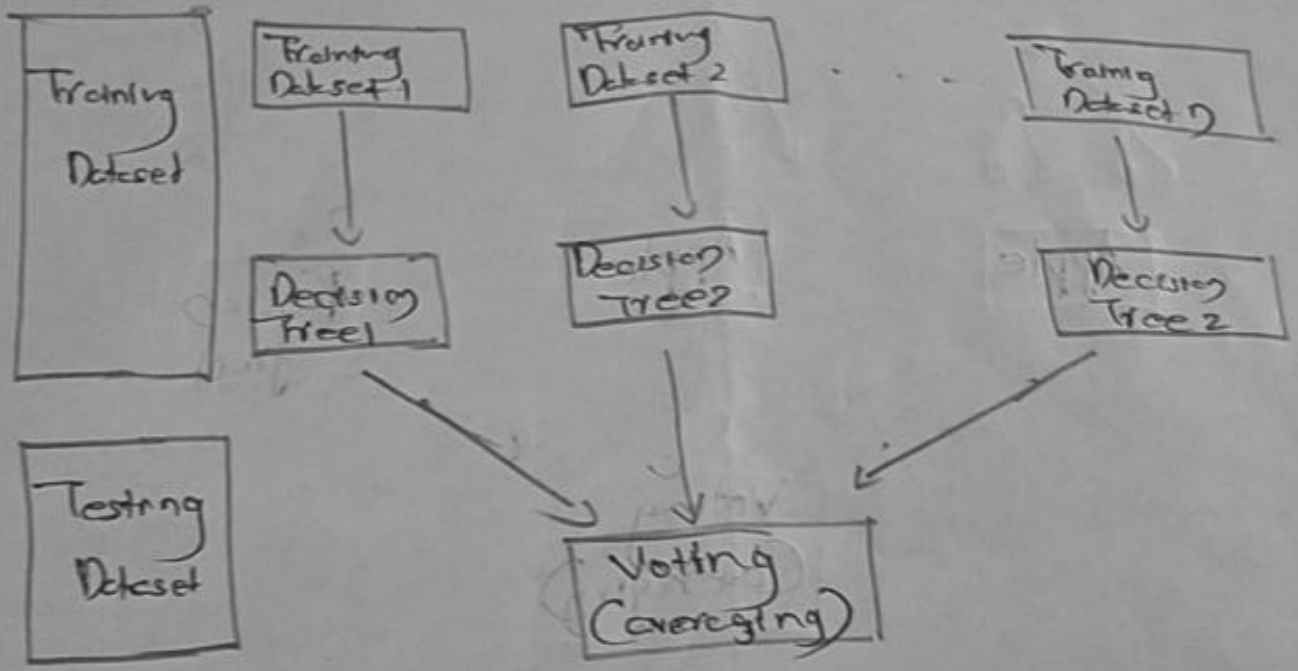
If data set having 60% of relevent and 40% of non relevent information

then,

$$\text{Entropy}(S) = -\left(\frac{60}{100}\right) \log_2 \left(\frac{60}{100}\right) + -\left(\frac{40}{100}\right)$$

$$= -0.6 \log 0.6 + -0.4 \log$$

6. Random Forest

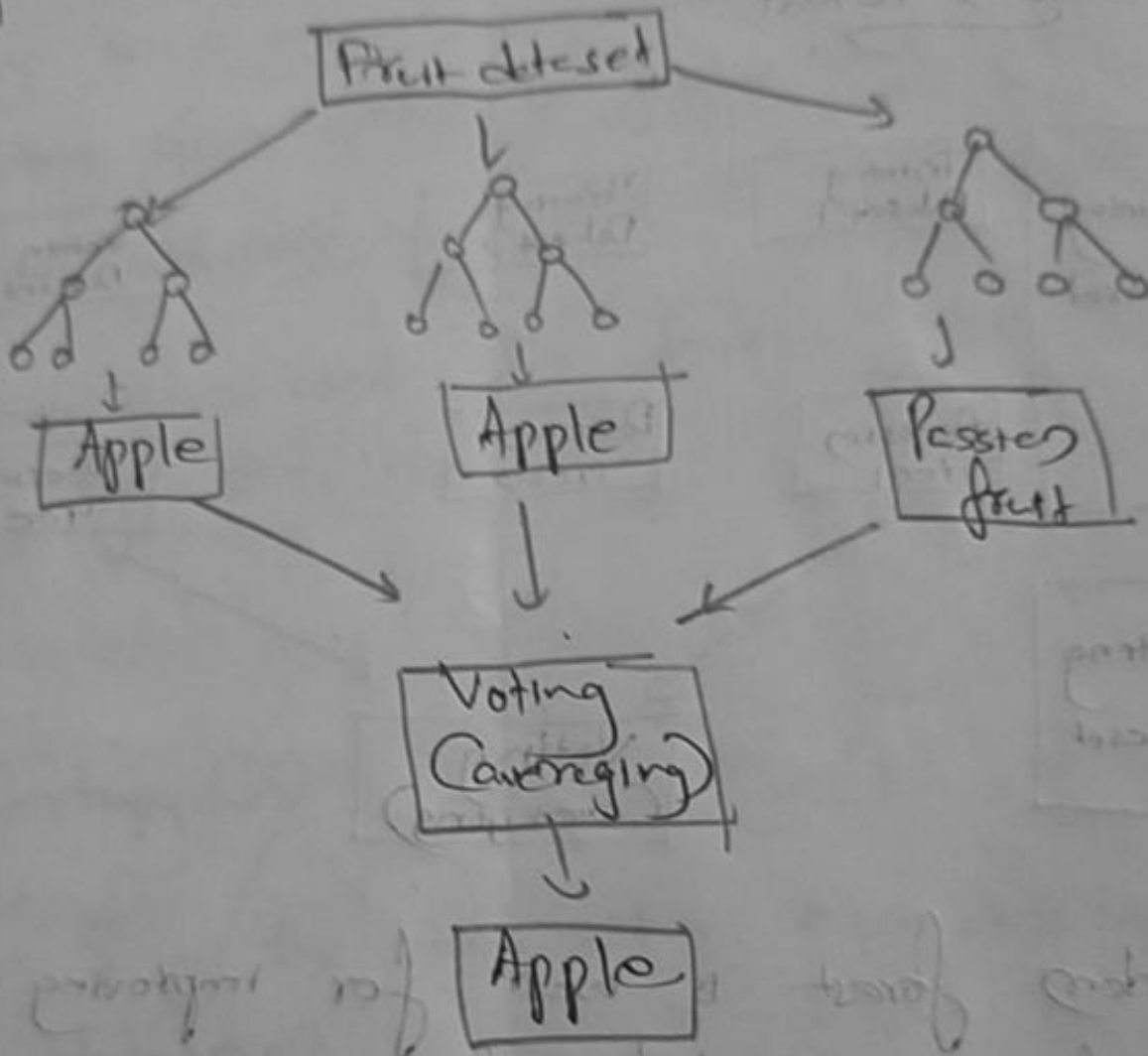


→ Random forest is used for improving the classification model performance

→ Random forest ~~actually~~ actually does is splitting ~~upof~~ of ~~Testing~~ ~~d~~ training data set on n no: of bootstrap sampling blocks

→ The Corresponding decision trees is constru for each Training data subsets and the

Example


Fruit dataset
Apple
Apple
Passion fruit
Voting (averaging)
Apple